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Rotational CARS thermometry in diffusion flames: On the influence of nitrogen spectral line-broadening by CH_4 and H_2

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Abstract

Rotational CARS N_2 thermometry has been developed for improved accuracy in spatial flame regions with high concentrations of either H_2 or CH_4, which are typical conditions on the fuel side of diffusion flames. In initial validation experiments up to 700 K, rotational CARS measurements were performed in binary mixtures of H_2 and N_2. Subsequent spectral evaluation with commonly used N_2–N_2 Raman line widths clearly underestimated the real temperatures. Inclusion of recently calculated N_2–H_2 spectral line-broadening coefficients weighted by the concentration of hydrogen resulted in CARS temperatures in much better agreement with the true temperatures. Based on this validation work, improved temperature accuracy was achieved in flame measurements on the fuel side of a hydrogen diffusion flame by probing nitrogen simultaneously with the detection of the hydrogen S_0-transition at 354 cm^{-1}, and weighting the species specific Raman line widths with the information of relative concentration. The validation experiments were repeated for binary mixtures of nitrogen and methane up to 500 K. It was shown that N_2–N_2 Raman line widths gave reproducible temperatures independent of methane concentration in binary mixtures with nitrogen, thus no explicit knowledge on N_2–CH_4 Raman line widths is needed for this temperature region. In general, the impact of Raman line widths for accurate rotational CARS thermometry is demonstrated.
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Keywords: Rotational CARS thermometry; Non-linear spectroscopy; Raman line widths; Diffusion flame

1. Introduction

Coherent anti-Stokes Raman spectroscopy (CARS) has been developed over the last decades to be an efficient tool for thermometry and sometimes also for the evaluation of relative species concentrations in combustion processes [1]. CARS is a point-wise technique and best adapted to measurements of small molecules, such as diatomics and triatomics. Since such molecules are present in high concentrations within combustion processes, a natural choice is to probe N_2, O_2, H_2, CO, CO_2, and H_2O. For CARS thermometry, the main focus has been on the ability of probing N_2. Temperature evaluation of an experimental spectrum is generally made by spectral fitting using a library with theoretical spectra calculated at different temperatures for the actual measurement conditions and experimental setup.

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CARS spectral shapes are very temperature sensitive, which is mainly due to temperature dependent changes in the Boltzmann thermal population distribution, for rotational CARS mainly of the rotational states in the ground level vibrational state. The second most important parameter in a rotational CARS spectrum contributing to the temperature sensitivity is the species specific Raman line widths, which are dependent on both temperature and rotational quantum number J [2]. For measurements in pure N_2 gas, only self-broadening of nitrogen spectral lines is considered, i.e. N_2–N_2 Raman line widths, whereas for measurements where nitrogen is probed in a mixture of gases, nitrogen spectral line-broadening from all present species should be taken into account. In such cases, the species specific Raman line widths should be considered in the theoretical model weighted by the species concentration. Line width data for different collision partners to N_2 are available from literature; N_2–H_2O [3], N_2–CO_2 [4], N_2–O_2 [5], and N_2–CO [6].

In rotational CARS measurements applied to sooting flames in 2005 [7], all important Raman line widths were included in the theoretical analysis except N_2–H_2 Raman line widths, which were still not investigated. The apparent lack of such data initiated experimental as well as theoretical work on the N_2–H_2 system [8]. It was found that the J-dependence of the spectral line-broadening coefficients for this system was very different from the coefficients of self-broadened nitrogen, and the implementation of these coefficients into the theoretical code systematically increased the evaluated temperatures by several tens of Kelvin [9]. To validate these behaviors, cell measurements in binary mixtures of H_2 and N_2 were performed, and it was observed that the evaluated temperatures clearly underestimated the real temperatures if the N_2–H_2 Raman line widths were not considered [10]. Instead with implemented N_2–H_2 Raman line widths weighted by the H_2-concentration, the temperature evaluation was significantly improved.

In this work, it is demonstrated how the information from validation experiments in binary mixtures of N_2 and H_2 can be used for improved thermometry on the fuel side of hydrogen diffusion flames. It is done by simultaneous detection of the rotational CARS spectrum of N_2 and the first rotational CARS spectral line of H_2, S_0 ($J \rightarrow J+2$) at 354 cm^{-1}, and the information can together with calibration cell measurements result in relative concentrations, which are used to improve the modelization of Raman line widths in the evaluation procedure. The proposed strategy is then demonstrated in the evaluation of a temperature profile in a hydrogen diffusion flame. In addition, validation cell experiments were performed for binary mixtures of N_2 and CH_4 up to 500 K. For these mixtures no significant change was observed in evaluated temperatures for increasing CH_4-concentrations using pure N_2–N_2 Raman line widths in the evaluation. This implies that N_2–CH_4 Raman line widths are very similar to N_2–N_2 Raman line widths, which simplifies temperature evaluation on the fuel side of methane diffusion flames. Based on the present results, efficient strategies for accurate temperature evaluation using rotational CARS are demonstrated.

2. Theoretical considerations

The general theory for CARS has extensively been described elsewhere [1], and for a detailed description regarding DB-RCARS (dual-broadband rotational CARS) readers are referred to [11] and references therein. Instead here the fundamentals of using the technique for gas-phase thermometry are briefly discussed. CARS is a non-linear optical technique based on a four-wave-mixing principle, where three laser beams (pump, Stokes and probe), which are focused and directed to a common point of intersection, are converted into a fourth through an interaction with molecular Raman transitions in a gas-phase medium. In the dual-broadband approach, the pump and Stokes beams originate from a broadband dye laser and the probed molecules are coherently excited when the energy difference between the incoming photons is equal to the Raman modes of the specific molecules. The probe beam, most often provided by a narrowband laser, is scattered from the induced states as anti-Stokes shifted coherent light (i.e. the CARS signal), which complete the four-wave mixing process. For pure rotational CARS these transitions take place within the rotational energy levels of the probed molecules, and each constituent has a species specific spectrum. Since many diatomic and other small sized molecules have a rotational constant in the same range, they are detected within the same spectral region (if the Raman cross section is strong enough and the concentration is on the order of a few percent). Hence, in a recorded rotational CARS spectrum with spectral contributions from more than one species, the signal strength of the constituents can be used to derive the relative concentration of these species.

The appearance of the overall rotational CARS spectral envelope is linked to the thermal population distribution of the rotational states from which temperature can be extracted. In the temperature evaluation procedure, a library of theoretical spectra generated in temperature steps relevant for the actual measurement situation, is fitted to the recorded experimental spectrum. By a Levenberg–Marquardt non-linear interpolating algorithm, temperature is deduced by minimizing the sum-of-squares (SSQ) of the residuals between

the recorded spectrum and the theoretical spectra in the library. The procedure implicates calculation of accurate theoretical spectra, which requires adequate information about the molecular system (molecular constants, Raman line widths) as well as information about experimental operating conditions (instrument function, spectral dispersion). Among these, the most critical for thermometry is the species specific Raman line widths, which describe how the spectral lines of the probed molecule get perturbed by collisions from other molecules present in the probe volume. For applied temperature measurements within combustion systems, nitrogen is the main probed molecule due to its high content in air and by acting as an inert species in the combustion process. Thus in a combustion environment probing nitrogen, collisions from all other present molecules influence the spectral line-broadening.

In Fig. 1, the Raman line widths from N_2–N_2 [12] and N_2–H_2 [8] are displayed for two temperatures; 300 and 500 K. The spectral line-broadening coefficients are plotted within a range of rotational quantum numbers covering more than 99% of the population for the nitrogen molecule at the particular temperature. At each temperature, it can be seen how the *J*-dependence of the coefficients from the collision partners N_2 and H_2 are clearly different, which consequently influence the total shape of the rotational CARS spectral envelope. In the calculations of a theoretical CARS spectrum to be used for the analysis of temperature from a measurement, the spectral line-broadening coefficients are weighted by the concentration of each species in the probe volume. If there is a limited knowledge on these coefficients, or if the species concentrations in the gas are uncertain, this may lead to incorrect modelling of the theoretical spectral shapes. Since the influence on the spectral shape from the Raman line widths are not decoupled from the influence from the Boltzmann population distribution, uncertainties in the Raman line widths will influence the temperature accuracy of the technique. The impact of the differences in the Raman line widths in Fig. 1 will be presented in the section Results and Discussion.

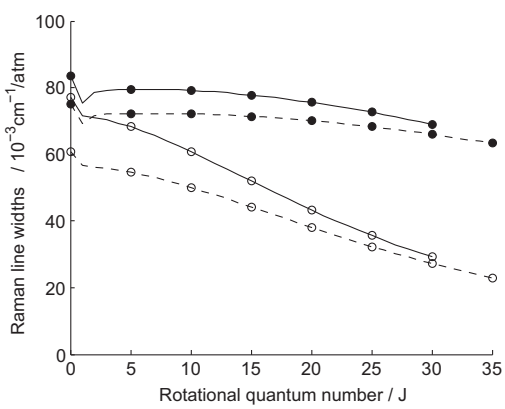

Fig. 1. Raman line widths (FWHM) N_2–N_2 [12] (open circles) and N_2–H_2 [8] (filled circles) shown at two temperatures, 500 K (solid), and 700 K (dashed).

3. Experimental procedure

A simplified scheme of the experimental setup is depicted in Fig. 2. A frequency-doubled injection-seeded Nd:YAG laser (Quantel YG:981E) producing spectrally narrow single-mode light (FWHM <0.01 cm^{-1}) at 532 nm, is pumping a dye laser and ~8% is split off to be used as the pump beam, ω_3, in the CARS process. The pump (ω_1) and Stokes (ω_2) beams are provided by a dye laser (Quantel TDL 90), prepared with Rhodamine (590, 610) in a solvent with methanol, to produce broadband light centred at 593 nm with a spectral emission width of ~10 nm (FWHM). The particular dye solution was mainly chosen to avoid a smeared vibrational CARS signal in gas mixtures with methane [13,14]. The pulses are generated at a repetition rate of 10 Hz with a temporal length of ~8 ns, and have typical pulse energies for the flame measurements of 20 mJ at 532 nm and 2 × 40 mJ at 593 nm. The laser beams are aligned in a planar BOXCARS phase-matching configuration [1] and arranged with a separation distance of 12 mm at the focusing lens ($f = 300$ mm). This arrangement results in a probe volume with an approximate length of 1–2 mm and width of 0.1 mm. After the re-collimating lens ($f = 300$ mm), the generated CARS beam is propagating nearly superimposed with the ω_1 red beam. To spectrally filter the CARS beam, dichroic mirrors and short-pass filters are inserted in the optical path together with several apertures to remove potential spectral interference from stray light, originating from the probe beam. However for the calibration measurements of this work, with spurious light encountered from reflections in the cell glass windows, the suppression of

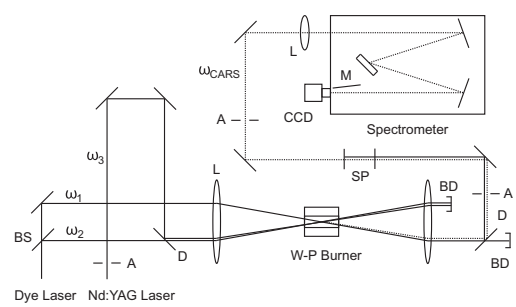

Fig. 2. Experimental setup for dual-broadband rotational CARS. A (aperture), BS (beam splitter), D (dichroic mirror), L (lens), W–P (Wolfhard–Parker), BD (beam dump), SP (short-pass filter), and M (mirror).

stray light from the probe beam was improved by angle tuning a short-wave-pass filter (Semrock, Razor edge) [15].

An $f = 150$ mm lens is focusing the CARS beam on the entrance slit of a 1-m spectrometer of Czerny-Turner type with a holographic grating (Newport Corporation). The DB-RCARS spectrum is recorded using a back-illuminated unintensified CCD camera (Princeton Instruments) having 1100×330 pixels with a size of 24×24 μm. The spectral dispersion at the exit of the spectrometer is 0.23 cm^{-1}/pixel and the limited spectral range of detection (~250 cm^{-1}) was optimized for the nitrogen rotational CARS spectral envelope. Therefore, to simultaneously detect the S_0 hydrogen spectral line at 354 cm^{-1}, a mirror (M) was inserted on the right side of the signal beam path to reflect the dispersed light of this angle to be collected in the same spectral window. The reflecting mirror was carefully positioned such that the hydrogen peak was recorded at a spectral position to not interfere with the nitrogen spectrum. The instrument function of the detection system was optimized by a Voigt profile (Gaussian 0.3 cm^{-1}, Lorentzian 0.2 cm^{-1}). All spectra were divided by a spectrum recorded in a non-resonant gas of argon to compensate for the finite width of the broadband laser profile. This procedure also compensated for any difference in transmission characteristics of the spectral detection.

For the validation and calibration measurements, an aluminium cylindrical cell was used. It is equipped with anti-reflection coated quartz windows, and has a distance of 240 mm between the windows and an inner cell diameter of 28 mm. A standard heating tape (Hemi Heating AB), with an approximate maximum load of 150 W/m for heating up to 700 K, was wrapped around the cell. Three thermocouples of type K (Pentronic AB) were mounted along the central axis of the cell, over a distance of 60 mm, for control of the temperature.

For the flame measurements, a Wolfhard Parker burner was used. The burner is an exact replica of the one used at NIST in numerous investigations, see Ref. [16] and references therein. Hydrogen was flowed in the inner slot (length 40 mm, width 8 mm) at a flow rate of 2 l/min., and air at a flow rate of 8 l/min in the two outer slots, each with a length of 40 mm and width of 16 mm. The stability of the hydrogen diffusion flame was ensured through observation of the visible flame emission [17]. In the CARS measurements, the laser beams were propagating along the slots of the burner to achieve high spatial resolution in the direction where strong temperature gradients exist [18].

4. Results and discussion

In this section, firstly validation experiments are presented from investigated influence of nitrogen spectral line-broadening by H_2 and CH_4 on evaluated temperatures. This information is essential for the temperature accuracy of rotational CARS nitrogen thermometry in regions with high concentrations of hydrogen and methane. Secondly, calibration measurements describe how the concentration ratio between hydrogen and nitrogen is related to the intensity ratio between the S_0 hydrogen spectral peak and the rotational CARS signal of nitrogen. Finally, it is shown how these validation and calibration measurements are necessary in order to make corrections to rotational CARS temperature evaluations on the fuel side of a hydrogen diffusion flame.

4.1. Validation experiments on the importance of spectral line-broadening coefficients

Binary mixtures of hydrogen and nitrogen were encased in a cell, where the hydrogen concentration was varied between 0% and 90% in steps of 10% at three temperatures; 300, 500, and 700 K. In Fig. 3, the results for the 500 K-case are shown, while the complete results are presented in Ref. 10. The triangles show how the real temperature is clearly underestimated if probing nitrogen in a binary mixture with hydrogen, and only using self-broadened N_2–N_2 Raman line widths (see Fig. 1) in the analysis. Moreover, it can be observed that the underestimation increases almost linearly with increasing hydrogen concentration. By also implementing the N_2–H_2 Raman line widths presented in Fig. 1 properly weighted by the concentration, the evaluated temperatures are much closer to the real tempera-

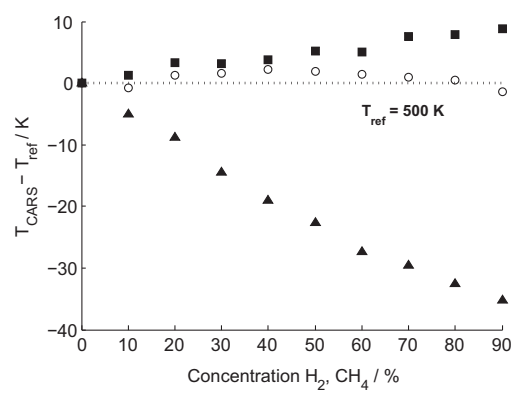

Fig. 3. Rotational CARS temperatures evaluated with self-broadened N_2–N_2 Raman line widths (triangles) only and combined with N_2–H_2 Raman line widths (squares) displayed relative the reference temperature 500 K as a function of hydrogen concentration in the cell. In similar validation experiment with binary mixtures of nitrogen and methane, CARS temperatures evaluated with pure self-broadened N_2–N_2 Raman line widths (open circles) is presented.

tures, shown as squares in Fig. 3. However, the result is even a slight overestimation.

In a second series of measurements, binary mixtures of nitrogen and methane were investigated for temperatures up to 500 K, and the results for evaluations made with coefficients of self-broadened nitrogen at 500 K are shown in Fig. 3 as circles. Here the deviation is small between the evaluated CARS temperature and the reference temperature even at a higher concentration of methane in the cell. This behaviour implies that the nitrogen spectral line-broadening from collisions with methane is very similar to the one of collisions with nitrogen itself, which supports the results from the only (to the authors knowledge) reported investigation on N_2–CH_4 Raman line widths [19]. This proves that in the same temperature region with rotational CARS thermometry on nitrogen in mixtures with methane, the Raman line widths from N_2–CH_4 can be approximated with the Raman line widths from N_2–N_2 with negligible loss in accuracy. The same behaviour of temperature independence in evaluated nitrogen temperature for increasing methane concentration was observed for lower temperatures.

It should be noted that the measurements leading to the results in Fig. 3 were repeated several times, all showing the same results. Based on this observation and the fact that the data in Fig. 3 deviate little from a fitted curve to the data points we estimate the uncertainty in each point to be around 3 K. The temperature T_{ref} for the various experiments showed differences in the range of 0–3% from the absolute thermocouple temperatures although the difference in most cases was below 1% [10]. As the resonant part of the CARS spectrum was so much stronger than the non-resonant part for all investigated mixtures, the non-resonant background and uncertainties in the instrument function had negligible impact on the temperature accuracy.

The results shown in Fig. 3 for N_2–H_2 Raman line widths at 500 K, were also performed at 300 and 700 K, and they show similar behaviour, i.e. that self-broadened N_2–N_2 Raman line widths underestimates the evaluated temperatures and that the implementation of N_2–H_2 Raman line widths weighted by their concentrations leads to a slight overestimation [10].

In a more general discussion the presented experiments demonstrate an empirical way of deriving line-broadening data. For example, the N_2–H_2 Raman line widths could be tuned to result in a perfect agreement between the CARS temperatures and the real temperatures in Fig. 3 for all hydrogen concentrations and could be supplementary to inverse Raman measurements to extract N_2–H_2 Raman line widths [8]. However, since the total modelization of nitrogen spectral line-broadening consists of a linear combination between Raman line widths from both N_2–N_2 and N_2–H_2, the procedure is assuming the coefficients of self-broadened nitrogen to be correct.

The remaining part of this section Results and Discussion will only treat N_2–H_2 Raman line widths and their use for flame thermometry.

4.2. Calibration experiments for relative hydrogen/nitrogen concentration

When applying the rotational CARS technique to a hydrogen diffusion flame, the results in Fig. 3 shows that knowledge about the hydrogen concentration is necessary for accurate temperature evaluation. Thus a way of detecting hydrogen is demanded to estimate the hydrogen concentration and thereby to be able to use appropriately weighted Raman line widths in the temperature evaluation. In fact the first rotational Raman hydrogen transition, S_0 at 354 cm^{-1}, offers such a possibility for detection.

The calibration of concentration was performed, similar to the validation experiments, in binary mixtures of hydrogen and nitrogen encased in a cell, where the temperatures were varied in the range of 300–700 K in steps of 100 K, and the hydrogen concentration was varied between 0% and 90% in steps of 10%. In Fig. 4, rotational CARS spectra at 500 K are presented for four different relative concentrations of hydrogen and nitrogen, with 60–90% hydrogen. All spectra have normalised intensity at the single hydrogen spectral peak. It can be seen how the hydrogen S_0 spectral line at Raman shift 354 cm^{-1} instead appears at about 270 cm^{-1} by the special mirror arrangement in the experimental setup to collect this angle of the dispersed light. The rotational CARS spectrum from nitrogen shows its characteristic signature of equidistant spectral lines with an intensity ratio of 1:4 between adjacent spectral

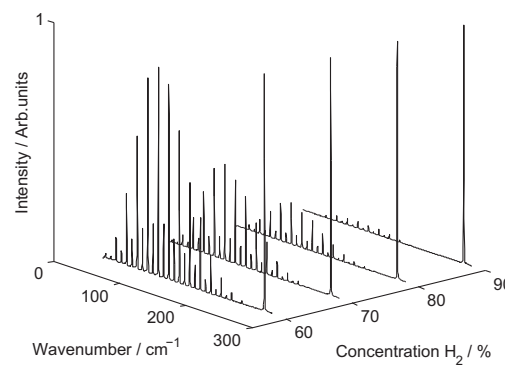

Fig. 4. Rotational CARS spectra from simultaneously detected hydrogen (S_0) and nitrogen in binary mixtures of N_2 and H_2 with 60–90% H_2. The spectra are averaged from 300 single shots and normalized to the hydrogen peak.

lines [2]. The spectral shape is strongly dependent on temperature, where a higher temperature leads to a broader spectrum with an intensity distribution displaced towards higher Raman shifts [2].

The peak intensity ratios between H_2 (S_0) and the N_2 spectral lines as a function of concentration were established for all the investigated temperatures, and to improve the accuracy several peak ratios were used in the calibration procedure. The CARS signal strength is dependent on the square of the species concentration, and as can be seen in Fig. 4, there is a strong significance in ratio between hydrogen and nitrogen as a function of relative concentration. Also, there is a strong temperature dependence of the intensity ratios between H_2 (S_0) and the N_2 lines, which is related to changes in thermal population distribution versus temperature for both H_2 and N_2.

4.3. Hydrogen diffusion flame measurements

Temperatures were measured on the fuel side of a hydrogen/air diffusion flame, provided on a Wolfhard–Parker (WP) slot burner [16] by probing the nitrogen molecules diffusing into the fuel region from the air flow supplied in the outer slots. Measurements were performed by scanning the burner perpendicularly to the burner slots, from one flame front to the other in steps of 0.5 mm at a height of 1 mm above burner surface. According to the calibration measurements in Section 4.2, simultaneous detection of the S_0 hydrogen spectral line and the entire signal from nitrogen was performed. To extract the actual relative concentration between hydrogen and nitrogen in the flame, the ratios between the S_0 hydrogen spectral peak and the S-branch nitrogen spectral peaks were compared with the results obtained from the calibration measurements discussed in Section 4.2.

The evaluated temperatures are shown in Fig. 5, where for each position the evaluated temperature is the average of 300 single-shot spectra. The first analysis of the rotational CARS data was made utilizing N_2–N_2 Raman line widths at all spatial positions, which is represented as open circles. In the second and more appropriate analysis, the flame temperatures were obtained iteratively, by first estimating the temperature from evaluations made with pure self-broadened N_2–N_2 Raman line widths to decide in what temperature interval the relative concentrations should be extracted, and then to further correct these temperatures from evaluations made with weighted Raman line widths according to the composition of the gas. These data are shown as filled circles and the evaluated temperatures are higher at all spatial locations. The difference between the two sets of temperature data is about 50 K for all positions, which can be explained by a combination of two effects. At the centre, the difference is dominated by a high concentration of

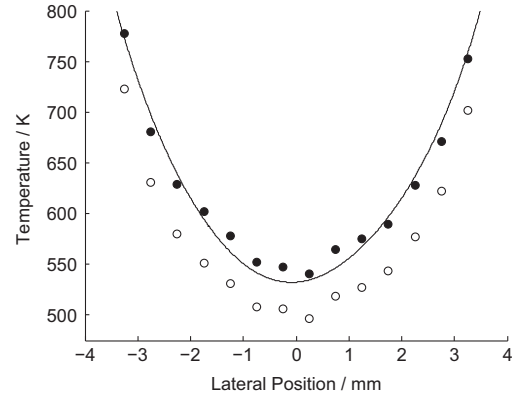

Fig. 5. Temperatures from CARS measurements in a hydrogen diffusion flame provided on Wolfhard–Parker (WP) slot burner at 1 mm height above burner edge. Evaluations were made with two different sets of Raman line widths, either with N_2–N_2 (open circles) or with weighted N_2–N_2/N_2–H_2 (filled circles) according to the extracted relative concentration. The line shows results from a fitted polynomial of thermocouple temperature data at the same spatial coordinates in the flame.

hydrogen estimated to about 90–95%, and the underestimation in evaluated temperatures from the Raman line widths of pure self-broadened nitrogen are about the same as for the validation measurements in the cell presented in Fig. 2. Towards the edges of the burner slot providing the fuel, at around 3 mm, the temperature is higher, which should mean a greater difference in evaluated temperatures between the two sets of Raman line widths if the relative concentration would have been constant according to Ref. 10. But combined with a lower concentration of hydrogen estimated to about 55–60%, the difference in evaluated temperatures between these sets is roughly unchanged.

To validate the evaluated CARS temperature data for the more elaborate analysis involving N_2–H_2 Raman line widths, thermocouple temperature measurements were performed at the same positions as the CARS measurements. A thermocouple of type K (Pt/Pt-13% Rh), with a thickness of 80 μm, was inserted at the same height parallel to the burner slots to minimize effects of heat conduction along the thermocouple leads. The thermocouple data are represented in Fig. 5 as a solid line, where the individual temperatures have been replaced by fitted values of a higher order polynomial, for which the standard deviation from the residuals between the measured data and fitted values were about 7 K. This was done in order to clarify the differences between the thermocouple temperatures and the two evaluations of CARS temperatures. The data obtained from these measurements were not corrected for radiation losses which are only significant at much higher temperatures [16]. In Fig. 5, it can be seen

how the CARS temperatures with implemented N_2–H_2 Raman line widths are in much better agreement with the thermocouple temperatures than the evaluations made with pure self-broadened N_2–N_2 Raman line widths. Thus, qualified by the thermocouple temperatures, it is verified that the accuracy of the rotational CARS nitrogen thermometry in an environment containing large amounts of hydrogen is improved by the implementation of N_2–H_2 Raman line widths.

The evaluated temperatures with the weighted N_2–N_2/N_2–H_2 Raman line widths (solid circles in Fig. 5) seem to be slightly overestimated in comparison with the thermocouple temperatures in the centre of the flame, where the majority of the collisions are from hydrogen. However, the difference may not be significant given the uncertainties present in the analysis. One of the uncertainties is the correlation in vertical position between the thermocouple and CARS measurements. We estimate this uncertainty to be up to 0.2 mm, which was found to correspond to a temperature difference of 12 K. Further, it should be noted that for this work no consideration is taken to nitrogen spectral line-broadening from water, which should be added in a complete description of molecular collisions influencing the nitrogen spectral shape. Such implementation of N_2–H_2O Raman line widths would systematically raise the evaluated temperatures slightly according to the *J*-dependence of these coefficients [7]. This is verified with tests for measurement points at the edges of the burner slot providing the fuel, where implemented coefficients of this kind weighted with a water concentration of 10% raises the temperature about 4 K. Moreover, there is the general uncertainty in rotational CARS nitrogen thermometry, which often is considered to be up to 2–3% of the absolute temperature.

In this work, pure rotational CARS nitrogen thermometry has been suggested as an interesting technique to be applied at the fuel side of hydrogen and methane diffusion flames, where nitrogen that has diffused into the fuel region is probed. Vibrational CARS thermometry is another possible diagnostic for such situations. However, vibrational CARS H_2 thermometry at these low temperatures has to rely on the analysis of a few spectral lines only and as a result from a recent study spectra must be properly modelled by inclusion of Herman-Wallis factors [20]. Vibrational CARS CH_4 thermometry also relies on the detection of a few spectral lines and additionally demands a high spectral resolution, which is comprehensively discussed in Ref. [21]. Moreover, there is a large number of CARS techniques that can be used for extraction of temperature and major species concentrations, for example simultaneous vibrational and rotational CARS [22,23], and dual-pump CARS [23,24]. Also, there are novel methods for CARS on very short time scales, which are highly interesting because of their ability of being insensitive to collisions [25]. The here presented rotational CARS technique is an interesting alternative for thermometry because of its high temperature sensitivity at the low temperatures persisting on the fuel side of diffusion flames and the well investigated dependence on nitrogen spectral line-broadening from H_2 and CH_4.

5. Conclusions

1. When probing nitrogen using rotational CARS thermometry in an environment of ambient hydrogen, the N_2–H_2 Raman line widths should be included in the calculation of theoretical spectra. If neglected, the evaluated CARS temperatures clearly underestimate the real temperature.
2. In order to extract rotational CARS temperatures with high accuracy on the fuel side of a hydrogen diffusion flame, knowledge on the hydrogen concentration is needed. For this reason, detection of the first hydrogen spectral line S_0 at 354 cm^{-1} together with the nitrogen spectrum is demonstrated. Calibration measurements in the range of 300–700 K, were performed to establish the signal ratio between H_2 (S_0) and N_2 spectral lines for different concentrations.
3. For the rotational CARS measurements performed on the fuel side of hydrogen diffusion flames, the extracted relative nitrogen/hydrogen concentration was used as input data for the evaluation so that N_2–H_2 and N_2–N_2 Raman line widths could be properly weighted. By this approach, good agreement was obtained with thermocouple temperatures.
4. When probing nitrogen using rotational CARS thermometry in a mixture with varying concentrations of methane, there is no significant change in evaluated temperature although only N_2–N_2 Raman line widths is used. Thus no explicit N_2–CH_4 Raman line widths must be used in the spectral evaluation procedure, instead the self-broadened N_2–N_2 Raman line widths is a good approximation for the nitrogen spectral line-broadening of CH_4. This result simplifies rotational CARS thermometry on the fuel side of methane diffusion flames.
5. Validation experiments in binary mixtures using rotational CARS has in this work demonstrated its potential in extracting information about nitrogen spectral line-broadening in an empirical way.
6. As a result of the present work a more general conclusion can be drawn that knowledge of line-broadening coefficients is important for accurate rotational CARS thermometry.

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